



StepMania

A Video Game Analysis by Naomi Kendall

THE BASICS

Game Name: StepMania

Platform: PC

Genre: Action, Rhythm Game, Musical Game

Theme: Music & Songs, Rhythm, Visual Mimicking, Directional Arrow

Tone: Challenged, Accomplished, Addictive

Age: 6+

OVERVIEW

StepMania is a 2D fast-paced single-player (with multiplayer capabilities) rhythm game that originally aimed to imitate the popular arcade game- Dance Dance Revolution, but has since evolved now into a more extendable rhythm game engine, capable of supporting various different types of rhythm games in its program.

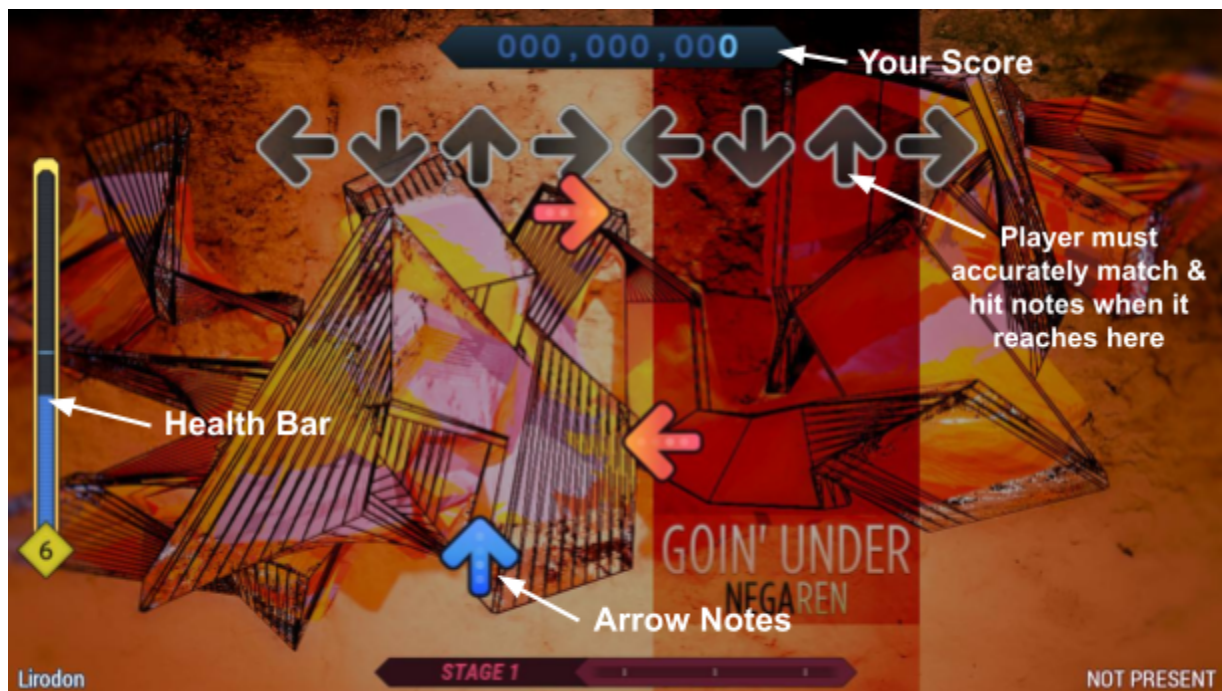
The aim of the game is to hit the arrow icons (known as notes) when they reach the top of the screen, as accurately and on the beat as possible.

HUD & UI

Song Menu UI



In-Game HUD



MDA MECHANICS

Notes

Arrow icons (known as notes) will fly in vertically across the screen from the bottom to the top in various patterns and BPMs (Beats Per Minute) in sync with the song chart being played.

Players can hit these notes in four different directions indicated by the arrow icons; left, right, up & down.

These notes also may appear in different colours. Each colour depicts a different beat the note falls on (eg. Red 4th Note, Blue 8th Note..).

There are two different types of notes the player can come across:

Tap Notes

Press the arrow when it reaches the top on the beat once.

Hold Notes

Press & hold the arrow when it reaches the top, and let go only when the stream finishes.

Both these types of notes may appear at different times or at the same time as each other, in which case the player will need to hit the notes at the same time.

Songs & Charts

In StepMania, players are required to import charted songs of their choice into the game from their website.

These songs have their own unique song chart, which includes patterns, notes, etc..that sync up to the music and follows the beat of the song.

Difficulty

There are five different difficulties to choose from -

Novice, **Easy**, **Medium**, **Hard** & **Expert**.

As the difficulty increases, the charts of the songs get harder, with more complex patterns, rhythms to follow, or fast streams of notes. (Not for the faint-hearted)

Judgment

While in-game, players will receive feedback comments (known as judgement) based on the timing accuracy of the last note they played.

There are six different types of judgement the player can get-

Flawless, **Perfect**, **Great**, **Good**, **Bad**, & **Miss**.

If a player continuously gets “Flawless”, “Perfect” or “Great” they will start a combo streak which goes towards their final player score of the song. However, receiving any other comment than the above will immediately break the combo streak.

Health Bar

While in-game, players have a health bar, which will decrease whenever the player plays notes too inaccurately, and increases if the player plays notes well.

If the player plays too inaccurately, and their health bar reaches 0, the song and the game will end, and the player will receive an F grade.

Player Score & Grade

When players play a song and get decent judgements (specifically Flawless, Perfect & Great) they will start earning points that go towards their final score for that song.

At the end of a song, the player will receive an evaluation of their performance on that song. That evaluation includes your final score, grade, and overall song percentage.

The final score the player receives, determines the overall percentage of the song 100% perfected. That percentage will go towards determining the grade assigned to the player as well. Grades serve as a visual indicator of a player's overall performance, grades that can be obtained are:

AAA (99%)

AA (95%)

A (90%)

B (80%)

C (70%)

D (50%)

F (49% and below/Fail Song)

MDA DYNAMICS

The main dynamics that are present while playing StepMania are:

- ❖ Needing to adopt a good ear for music beat & rhythm to become more precise with timing.
- ❖ Learning the most common patterns and sequence of arrows in song charts, and learning how to execute them.
- ❖ Familiarising with and getting used to hitting the notes on the screen using the keyboard keys or other controls, and learning the best ways to execute certain song movements to make one move to another feel more natural and less difficult to physically do.
- ❖ Becoming socially competitive with Stepmania's community, and friends/family, due to the scoring system, allowing players to compare scores and try to best each other.

MDA AESTHETICS

The player feels a great sense of Challenge and Submission from playing StepMania.

Challenge:

Is evoked from the game wanting the player to stay on the beat and hit the notes on time,

as accurately and precisely as they can. Because the best result is displayed for each of the song's they've played, they may feel compelled to keep beating their current score until they fully perfect the song (AAA 100%). The game also has multiple difficulties, so no matter the current skill ability of the player, they will always be given the opportunity to be challenged in the game.

Submission:

Is evoked from the repetition from practising a song over and over again, especially if a player wants to achieve mastery over a song's chart, and receive the best result possible. If they fail a song (maybe due to it being too difficult for their skill ability currently, or just unfamiliar patterns), they may feel compelled to keep grinding to try and improve their skills, and practise the song chart until they are able to pass the song.

PLAYER INPUT

Player Controls	Function
W/Up Arrow Key	Up Arrow Note/Navigate Menu
A/Left Arrow Key	Left Arrow Note/Navigate Menu
S/Down Arrow Key	Down Arrow Note/Navigate Menu
D/Right Arrow Key	Right Arrow Note/Navigate Menu
Esc	Open Pause Menu In-Game (Forfeit, Restart, Continue)
Space	Opens Settings Menu In-Game (Adjust Visuals, Arrow Speed, Toggle On/Off Gimmicks etc)/Confirm UI

Players press their keyboard arrow keys (or WASD keys) to use the corresponding arrow buttons on the screen.

PLAYER FEEDBACK

When the player accurately matches their movements to the onscreen arrows, the game rewards the player by giving them positive on screen comments on their performance (such as “Flawless!” and “Perfect”) and if done well continuously in a string of movements, will start to give the player a combo bonus, which goes towards their final score.

However,

When the player inaccurately matches their movements to the onscreen arrows, the game punishes the player by giving them negative on screen comments (such as “Miss” or “Bad”) and decreases their health bar, which if reaches zero will end the song, and the game will end abruptly. Any combo streak that was held, will be lost as well as soon as an inaccurate movement is made.

This feedback pushes the player to do the very best they can in order to keep themselves alive in the game and receive the more rewarding and encouraging positive comments.

SOUND

Music

Because of StepMania’s ability to import a massive array of different kinds of songs into its game, the songs and music of the game is never a constant genre. However all songs imported into StepMania will have some sort of distinct and pronounced beat or rhythm for the player to follow in order for it to be charted well, and playable.

ART STYLE

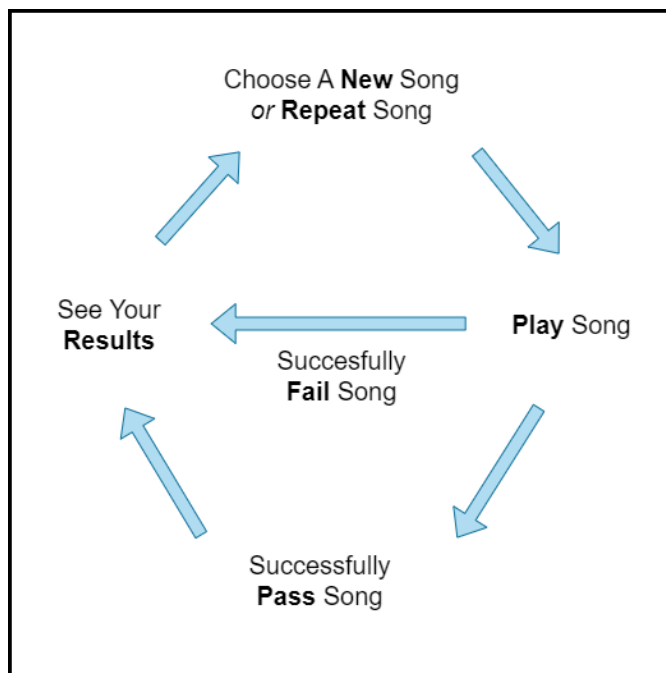
StepMania’s art style leans towards a retro arcade aesthetic, using many angled shapes, gradients and bright colours that jump out from the screen. The in-game background for the game constantly changes depending on the song being played, displaying the song’s custom background while you play, sometimes animated, sometimes static. The UI menu background take on an abstract shape design, with various different angled shapes.

STORY

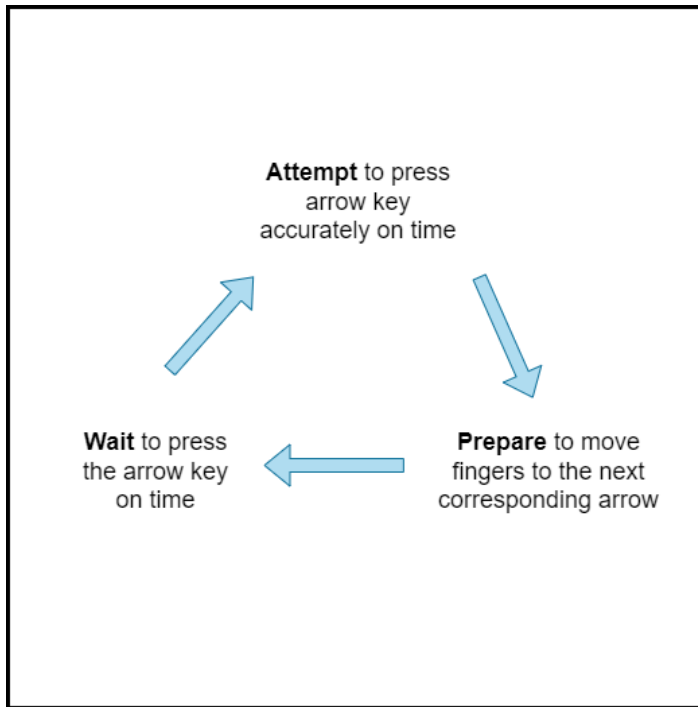
StepMania does not include any sort of narrative or storyline in its game, and it does not come up within the gameplay in any shape or form.

GAME LOOPS

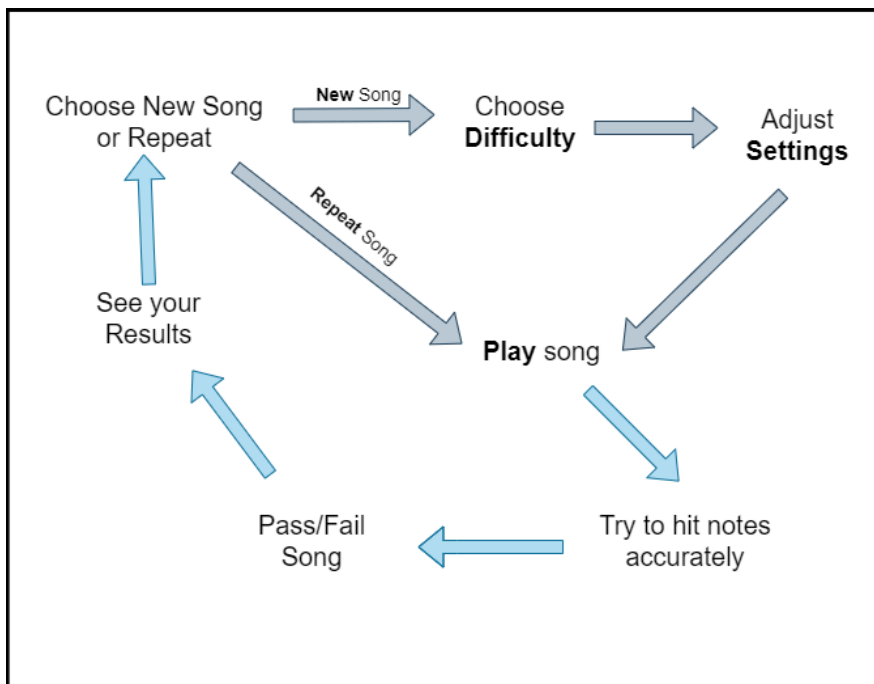
Core Game Loop



Moment-to-Moment Game Loop



Level Game Loop



Exit Points

Players at any point are able to pause the in-game state by pressing “Esc” and bring up the pause menu, which then gives the player the option to:

Continue

Unpauses the game state and gameplay continues off where it stopped

Restart

Restarts the song from the beginning, score points & health will be reset as well.

Forfeit

Exits the current song and returns the player to the song selection screen. Score points accumulated will be lost once you forfeit.

DEMOGRAPHIC

Because StepMania’s primary target audience is Dance Dance Revolution (DDR) players looking to practise the song charts through computer-based means, the majority of players playing StepMania will be found playing DDR or other games in the rhythm game genre, such as *Mai Mai*, *OSU!* or *Pump It Up (PIU)*.

Its core demographic consists of mostly adult men aged between 20 to 30 due to its inherently competitive and challenging nature.

StepMania’s demographic primarily uses the PC as their platform as this is the only platform the game is available on. However, some may use a unique combination of both the PC and external dance pads, setting the game up on a monitor and connecting the games controls to function from stepping on the pads.